

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Steroid Metabolic Process (GO:0008202)	0.15944633	56	3.767e-05	2.022e-01	HSD17B2:14 HSD11B1:107 CYP11A1:201 CYP51A1:375 HSD17B3:409 CYP21A2:412 CYP27A1:499
'De Novo' AMP Biosynthetic Process (GO:0000000)	-0.15294345	4	2.895e-01	1.980e-01	PFAS:128 PAICS:5819 ATIC:5819 ADS51:5819 NA NA NA
'De Novo' Post-Translational Protein Fold (GO:0000000)	0.09781599	15	1.898e-01	1.980e-01	SDF2L1:887 DNAJ13:1097 HSPA1A:1742 GAK:1896 EROI1A:2443 ENTDP5:6847 PTGES3:6847
2-Oxoglutarate Metabolic Process (GO:0000000)	0.15259091	13	6.685e-02	1.980e-01	GOT2:166 TAT:169 OGDHL:185 GOT1:484 L2GHOD:1540 DHI:6847 PHYH:6847
3'-UTR-mediated mRNA Destabilization (GO:0000000)	-0.02404704	11	7.825e-01	1.947e-01	DND1:1427 TRIM71:5819 RC3H1:5819 ZFP36L1:5819 ZFP36L2:5819 MOV10:5819 ZFP36:5819
3'-UTR-mediated mRNA Stabilization (GO:0000000)	0.07242310	11	4.057e-01	1.980e-01	MAPK14:1303 HNRNPCL:1409 HNRNPFO:1754 DAZL:2202 ZFP36:6847 TARDBP:6847 YBX3:6847
3'-Phosphoadenosine 5'-Phosphosulfate Me (GO:0000000)	0.24757905	8	1.532e-02	1.980e-01	SULT1A2:97 SULT1B1:964 SULT2B1:1030 SULT1A3:1405 SULT1E1:1680 PAPSS2:6847 TPST2:6847
5S Class rRNA Transcription By RNA Polym (GO:0000000)	-0.04055290	5	7.535e-01	1.980e-01	GTF3C2:5819 GTF3C3:5819 GTF3C4:5819 GTF3C5:5819 GTF3C6:5819 NA NA NA
7-Methylguanosine RNA Capping (GO:000945)	0.20867202	2	3.067e-01	1.980e-01	RNMT:569 CMTR2:6847 NA NA NA NA NA
7-Methylguanosine mRNA Capping (GO:00063)	0.20867202	2	3.067e-01	1.980e-01	RNMT:569 CMTR2:6847 NA NA NA NA NA
ADP Transport (GO:0015866)	-0.02679134	5	8.356e-01	1.980e-01	SLC25A24:470 SLC25A42:5819 SLC25A5:5819 SLC25A23:5819 SLC25A41:12017 NA NA
AMP Biosynthetic Process (GO:0006167)	-0.18086332	6	1.250e-01	1.980e-01	PFAS:128 APRT:855 PAICS:5819 ATIC:5819 ADS51:5819 NUDT2:5819 NA
ATF6-mediated Unfolded Protein Response (GO:0000000)	0.05763797	5	6.554e-01	1.980e-01	CREBZF:656 MBTPS1:6847 XBP1:6847 ATF6B:6847 DIT3:6847 NA NA
ATP Biosynthetic Process (GO:0006754)	0.02728418	13	7.335e-01	1.980e-01	TGFB1:780 ATP5F1C:1803 NUDT2:6847 VPS9D1:6847 ATP5PF:6847 ATP5PD:6847 ATP5PB:6847
ATP Metabolic Process (GO:0046034)	0.02856910	25	6.212e-01	1.980e-01	TGFB1:780 ENPP3:1053 BAD:1081 MYH6:1785 ATP5F1C:1803 PARG:2089 HSPA8:6847
ATP Synthesis Coupled Electron Transport (GO:0000000)	0.11562172	3	4.879e-01	1.980e-01	NDUF6:949 NDUF52:6847 NDUFY3:6847 NA NA NA NA
ATP Transport (GO:0015867)	-0.08235327	9	3.923e-01	1.980e-01	CALHM1:191 SLC25A24:470 SLC25A42:5819 SLC25A5:5819 SLC25A23:5819 CALHM3:5819 SLC17A9:5819
AV Node Cell To Bundle Of His Cell Commu (GO:0000000)	0.04574441	4	7.514e-01	1.980e-01	GJA5:2469 SCN5A:6847 CXADR:6847 GJC3:6847 NA NA NA
Arp2/3 Complex-Mediated Actin Nucleation (GO:0000000)	-0.09664430	10	2.900e-01	1.980e-01	WHAMM:727 ARPC5L:796 WASHC1:1166 ARPC1A:1319 ARPC5:5819 ARPC3:5819 ARPC2:5819
B Cell Activation (GO:0042113)	0.06407234	61	8.398e-02	1.980e-01	CD86:34 HDAC9:214 IFNK:316 IL4:679 CASP8:1036 CR2:1071 VCAM1:1120
B Cell Chemotaxis (GO:0035754)	0.05469964	5	6.617e-01	1.980e-01	CYP7B1:734 PIK3CD:6847 HSD3B7:6847 CXCL13:6847 GAS6:6847 NA NA
B Cell Differentiation (GO:0030183)	0.05402654	44	2.153e-01	1.980e-01	HDAC9:214 IFNK:316 IL4:679 CR2:1071 VCAM1:1120 TPTN2:1201 FLT3:3342
B Cell Homeostasis (GO:0001762)	0.03584862	7	7.426e-01	1.980e-01	TNFRSF13B:400 ME2FC2:1121 GAPT:6847 DCKO1:6847 NCKAP1:6847 TNFAIP3:6847 DOKK1:112293
B Cell Proliferation (GO:0042100)	0.07943161	15	2.870e-01	1.980e-01	IFNK:316 CR2:1071 ME2FC2:1121 PTPRC2:183 IL10:6847 CD70:6847 IFNE:6847
B Cell Receptor Signaling Pathway (GO:0000000)	0.07446488	26	1.890e-01	1.980e-01	SH2B2:544 CTLA4:865 CD38:1042 ME2FC2:1121 RFTN1:2025 PTPRC2:183 VAV3:2243
BMP Signaling Pathway (GO:0030505)	-0.04136061	42	3.542e-01	1.980e-01	LEFTY3:241 LEFTY4:198 GDF10:1012 FAM83G:1081 INHBE1:1119 SLC39A5:1205 SMAD5:1264
C-terminal Protein Amino Acid Modificat (GO:0000000)	-0.08813923	8	8.800e-01	1.980e-01	AGBL4:781 GLPD1:967 AGLB5:5819 AGTPBP1:5819 FOLH1:5819 NA NA NA
C-terminal Protein Deglutamylation (GO:0000000)	-0.13997713	4	3.323e-01	1.980e-01	AGBL4:781 AGLB5:5819 AGTPBP1:5819 FOLH1:5819 NA NA NA
C-terminal Protein Lysylation (GO:0000000)	-0.03478644	3	8.347e-01	1.980e-01	GLD1:967 ATG16L1:5819 ATG17:1087 NA NA NA
C4-dicarboxylate, Alpha-Beta T Cell Activati (GO:0000000)	0.11620399	10	2.033e-01	1.980e-01	SLCIA1:216 SLC25A12:618 SLC25A18:992 SLC13A2:1451 SLC13A1:6847 SLC13A5:6847 UCP2:6847
CD4-positive, Alpha-Beta T Cell Activity (GO:0000000)	-0.13020632	5	3.133e-01	1.980e-01	KNK7:147 STOM1:242 SNTF1:5819 TNFSF4:5819 RSDA2:5819 HMGCS1:5819 NA NA NA
CD40 Signaling Pathway (GO:0023035)	0.03632796	7	7.393e-01	1.980e-01	CD86:34 CD40LG:6847 ITGB1:6847 PHB2:6847 TNIP2:6847 ITGA5:6847 RNF31:6847
CENP-A Containing Chromatin Assembly (GO:0000000)	0.04377507	5	7.371e-01	1.980e-01	NASP:1553 M518A:6847 CENPL:6847 HUJRP:6847 NA NA NA NA
COPI Coating Of Golgi Vesicle (GO:003482)	-0.05684820	2	7.807e-01	1.980e-01	ARFGAP3:1036 GBF1:10202 NA NA NA NA NA
COPI-coated Vesicle Budding (GO:0005860)	-0.05684820	2	7.807e-01	1.980e-01	ARFGAP3:1036 GBF1:10202 NA NA NA NA NA
COPII Vesicle Coating (GO:0048208)	-0.03438544	19	6.561e-01	1.980e-01	KLHL12:475 TRAPPC12:1193 TRAPPC3:5819 TRAPPC2:5819 TRAPPC1:5819 TRAPPC5:5819
COPII-coated Vesicle Budding (GO:0001114)	-0.01150596	24	8.035e-01	1.980e-01	TBC1D20:311 KLHL12:475 SH2D2A:491 TRAPPC12:1193 TRAPPC3:5819 TRAPPC2:5819 TRAPPC1:5819
CRD-mediated mRNA Stabilization (GO:0070)	-0.19756073	5	1.261e-01	1.980e-01	IGF2BP1:496 IYB1:1195 DHX9:5819 PABPC1:5819 HNRNPD:5819 NA NA NA
Cdc42 Protein Signal Transduction (GO:0000000)	-0.04055702	6	7.308e-01	1.980e-01	SHTN1:5819 WAS:5819 RHOV:5819 NTN1:5819 RHOBTB1:5819 RHOTB2:5819 NA NA
DNA Alkylation (GO:0006305)	0.09262387	15	2.144e-01	1.980e-01	DNMT3B:31 GATAD2A:146 BAZ2A:343 HELLS:1649 KMT2E:6847 MTRR:6847 ATRX:6847

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_IQ21_AMPLICON	-0.29109711	27	1.661e-07	1.075e-03	ADAM15:45 PBXIP1:80 MTX1:85 RUSC1:89 RORC:177 FOPD5:188 THBS3:190
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	-0.14502570	100	5.637e-07	1.825e-03	COL20A1:30 HELLZ:88 MTG2:97 CDH2B:104 FAM217B:130 LAMA5:149 SLC0A41:210
WP_METAPATHWAY_BIOTRANSFORMATION_PHASE_I	0.13697150	92	5.790e-06	1.250e-02	SULT1A2:97 AKR1D11:177 MGS11:161 CYP27B1:176 COMT:179 CYP11A1:201 GSTK1:235
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.23916671	26	2.444e-05	3.957e-02	HSD17B2:14 HSD11B1:107 AKR1D1:117 COMT:179 CYP11A1:201 HSD17B3:409 CYP21A2:412
REACTOME_BIOLOGICAL_OXIDATIONS	0.09961207	123	1.503e-04	1.947e-01	ALDH1A1:73 SULT1A2:97 HPGDS:136 MGS1:171 CYP27B1:176 COMT:179 CYP11A1:201
REACTOME_METABOLISM_OF_STEROID_HORMONES	0.19383671	28	3.868e-04	2.784e-01	HSD17B2:14 HSD11B1:107 CYP11A1:201 HSD17B3:409 CYP21A2:412 SERPINA6:585 CYP17A1:488
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS_	0.10671382	95	3.319e-04	2.784e-01	SLC36A1:71 SLC36A2:135 SLC12A3:193 SLC1A1:216 SLC6A6:251 SLC7A6:272 SLC20A1:1883
REACTOME_METABOLISM_OF_LIPIDS	0.04463701	576	2.908e-04	2.784e-01	HSD17B2:14 SLC01B1:127 PGTG1:96 HSD11B1:107 AKR1D1:117 HPGDS:136 RORA:170
CAIRO_HEPATOBLASTOMA_DN	0.07405818	196	3.441e-04	2.784e-01	TMEM131L:35 AKR1D1:117 SLC46A3:125 CA2:152 SLC7A5:177 OGDHL:185 AGXT:186
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACT	0.07628784	180	4.324e-04	2.800e-01	IFNAR1:58 IL2R:62 TNF:87 TNFSF10:103 TNFSF15:112 IL1RAP:304 IFNK:316
NIKOLSKY_BREAST_CANCER_8Q23_Q24_AMPLICON	-0.09228699	111	7.998e-04	3.176e-01	PLEC:24 FAM83H:206 GPR20:254 TRAPP9:286 OLAH:293 SCRIB:348 EEF1D:409
REACTOME_ERYTHROCYTES_TAKE_UP_OXYGEN_AND	0.04346604	5	8.339e-04	3.176e-01	CA2:152 CA4:253 AQP1:762 SLC1A1:1060 RHAG:2146 NA NA
WP_Q20112_COPY_NUMBER_VARIATION_SYNDROME	0.06778672	219	5.733e-04	3.176e-01	SLC22A16:3 SLC36A1:71 SLC01B1:82 SLC10A6:83 SLC36A2:135 SLC12A3:193 SLC10A2:198
WP_STEROID_HORMONE_PRECURSOR_BIOSYNTHESI	0.17941272	30	6.742e-04	3.176e-01	CNNM4:163 ARID5A:424 CNNM3:508 KANSL3:580 COL2A1:605 DUSP2:677 FAHD2B:759
COPI-coated Vesicle Budding (GO:0005860)	-0.05684820	2	7.807e-01	1.980e-01	AKR1D1:117 CYP11A1:201 CYP21A2:412 CYP17A1:883 HSD3B2:1180 SRD5A2:2125 SRD5A1:6847
KEGG_ALANINE_ASPARTATE_AND_GLUTAMATE_MET	0.20027634	24	6.854e-04	3.176e-01	GOT2:166 AGXT:186 CAD:245 IL4I1:373 GLS2:427 GOT1:484 ASL:589
KEGG_CYSSTEINE_AND_METHIONINE_METABOLISM	0.21228245	21	7.602e-04	3.176e-01	DNMT3B:31 DNMT1:113 GOT2:166 TAT:169 IL4I1:373 LDHC:408 GOT1:484
NIKOLSKY_BREAST_CANCER_16Q24_AMPLICON	-0.06055546	248	1.075e-03	3.783e-01	FTSJ3:58 HOXB13:67 LRRC59:71 ITGBA:100 ENGASE1:19 EVPL:121 SNF8:139
BENPORATH_HS_WITH_H3K27ME3	-0.03573796	730	1.117e-03	3.810e-01	HSD11B1:107 CYP11A1:201 HSD17B3:409 CYP21A2:412 CYP17A1:883 HSD3B2:1180 CYP19A1:2015
REACTOME_GLUCCOCHONDIAC_BIOSYNTHESIS	0.36870547	6	1.762e-03	4.226e-01	CEMP10:0 BNC1:14 PLEC:24 SUSD4:34 ADAMTSL3:50 HOXB13:67 DACH2:75
WP_ALLOGRAFT_REJECTION	0.11271893	65	1.691e-03	4.226e-01	HSD11B1:107 CYP21A2:412 SERPINA6:585 CYP17A1:883 HSD3B2:1180 HSD11B2:6847 NA
WP_PROXIMAL_TUBULE_TRANSPORT	0.13658419	44	1.732e-03	4.226e-01	CD86:34 TNF:87 HLA-DQB1:137 FOXO3:324 ABCB1:397 IL1A:608 IL4:679
KEGG_SULFUR_METABOLISM	0.30192728	9	1.710e-03	4.226e-01	ATP6V0A4:43 SLC36A2:135 CA2:152 SLC1A1:216 CA4:253 ABCB1:397 SLC5A8:692
KEGG_TYPE_1_DIABETES_MELLITUS	0.18085786	25	1.753e-03	4.226e-01	SULT1A2:97 SULT1A4:370 SULT2B1:1030 SULT1A3:1405 SULT1E1:1680 CHST13:2118 SUOX:2221
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	-0.04988831	333	1.866e-03	4.316e-01	CD86:34 TNF:87 HLA-DQB1:137 FOXO3:324 ABCB1:397 IL1A:608 IL4:679
REACTOME_ARACHIDONIC_ACID_METABOLISM	0.14099083	40	2.044e-03	4.564e-01	CHRN2B:26 SHE4D CAMK2B:46 HTR1D:55 KCNH2:81 VBX2:87 ZBTB16:98
REACTOME_CYTOCHROME_P450_ARRANGED_BY_SUB	0.14731339	35	2.573e-03	5.375e-01	PTGR1:96 HPGDS:136 GGT1:470 GPX4:500 CYP1A1:528 GGT5:642 GPX1:786
KEGG_GRAFT_VERSUS_HOST_DISEASE	0.21133524	17	2.559e-03	5.375e-01	CYP27B1:176 CYP11A1:201 CYP51A1:375 CYP21A2:412 CYP27A1:499 CYP1A1:528 CYP7B1:734
PID_NFAT_TFPATHWAY	0.14661319	35	2.697e-03	5.458e-01	CD86:34 TNF:87 HLA-DQB1:137 IL1A:608 CD80:787 IFNG:1016 FASLG:1216
REACTOME_MEIOTIC_RECOMBINATION	-0.15454536	31	2.912e-03	5.714e-01	TNF:87 FOXO3:324 HGKA:353 EGR4:646 IL4:679 CTLA4:865 NFATC3:915
REACTOME_TNFS_BIND_THEIR_PHYSIOLOGICAL_R	-0.17720343	23	3.271e-03	6.186e-01	MRE11:406 ATM:434 D2K2:626 MHL1:818 H3-4:831 H2AF9:1058 TOP3A:1215
PYEON_HVP_POSITIVE_TUMORS_UP	0.09915087	73	3.436e-03	6.186e-01	TNFSF15:112 TNFRSF13B:400 TNFSF14:613 TNFSF78:608 TNFRSF9:836 FASLG:1216 LTA:1294
KEGG_ALLOGRAFT_REJECTION	-0.18454841	21	3.439e-03	6.186e-01	TASOR:41 TAF1:154 UHRF1:155 TRIM56:411 DTL:423 GLS2:427 NR1D2:478
LASTOWSKA_NEUROBLASTOMA_COPY_NUMBER_UP	-0.07130891	138	3.908e-03	6.214e-01	CD86:34 TNF:87 HLA-DQB1:137 IL4:679 CD80:787 IFNG:1016 IL2B:1171
BIOCARTA_HBX_PATHWAY	-0.36844491	5	4.327e-03	6.214e-01	MINK1:51 FTSJ3:58 LRRC59:71 NEURL4:200 CERN1:265 PAHB:270 DPH1:276
IVANOVA_HEMATOPOIESIS_MATURE_CELL	0.05934998	203	3.673e-03	6.214e-01	SHC1:353 CREB1:565 LAMTOR5:675 SOS1:955 PTX2B:5819 NA NA
REACTOME_TP53_REGULATES_TRANSCRIPTION_OF	-0.16105691	26	4.487e-03	6.214e-01	ST3GALS4:81 IL33:127 GONT1:129 TSPAN1:67 TM4SF4:195 COL1A1:209 CPD:223
					TP53BP2:38 ATM:434 CASP2:465 AIFM2:528 TP73:610 PERP:828 PIDD1:1085

DisGeNET Top pathways by non-permutation

Sepsis	0.05204946	368	6.779e-04	2.117e-01	CD86:34 ATP6V0A4:43 TRPV5:55 RELB:82 TNF:87 IL33:127 PLAUR:164
Steroid-sensitive nephrotic syndrome	0.32496165	9	7.370e-04	2.117e-01	TNF:87 PLA2G7:654 VEGFA:756 ACE:981 CXCL8:1051 IL2B:1171 HPSE:1492
Dermatitis, Atopic	0.05653255	316	7.939e-04	2.215e-01	RELB:82 ALDH1A1:73 TNF:87 DNMT1:113 IL33:127 HPGDS:136 FCGRT:149
Immune thrombocytopenic purpura	0.08466666	130	8.925e-04	2.421e-01	DNMT3B:31 CD86:34 TNF:87 TNFSF15:113 SLC13:27 GCGRT:149 GALT:173
Acantchytosis	0.29915179	10	1.056e-03	2.423e-01	XX:45 VPS13A:206 CAD:245 GATA:142 KEL:709 PANK2:969 KCNN4:2254
Bacterial Infections	0.07199530	178	9.740e-04	2.423e-01	TNF:87 SLC27A5:177 MPEG1:188 CPD:233 NF2L2:305 GDE1:356 TCL1A:383
Diarrhea	0.06205069	237	1.075e-03	2.423e-01	DNMT3B:31 SCN11A:37 TNF:87 AKR1D1:117 SLC12A:139 CP7IA:293 FOXP3:324
Female hypogonadism syndrome	-0.20257727	22	1.010e-03	2.423e-01	FGF8:316 ALN51:321 ATM:434 PROKR2:436 WDR11:615 GNRHR:620 CHD7:768

KeggReactome Top pathways by non-permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
OME_METABOLISM_OF_LIPIDS	0.06841396	650	5.966e-09	1.076e-05	ALB:40 CYP17A1:47 HSD11B1:49 FABP1:59 SPTLC3:67 ELOVL7:97 HEXB:98
TRANSPORT_OF_SMALL_MOLECULES	-0.05511916	623	4.302e-06	3.398e-03	TRPV1:4 ANOS1:17 ALB:40 SLC22A16:52 SLC38A4:57 SLC13A2:69 SLC6A6:70
C_MEDATED_TRANSMEMBRANE_TRAN	0.08431892	232	1.131e-05	3.398e-03	SLC22A16:52 SLC38A4:57 SLC13A2:69 SLC6A6:70 SLC26A3:82 SLC52:92 SLC36
ETABOLISM_OF_AMINO_ACIDS_AND_D	0.07528992	299	9.284e-06	3.398e-03	RPL23:5 RPS14:16 BCKDK:21 TPST1A:31 GLS2:36 SOD5:37 TAT:54
OME_MEIOTIC_RECOMBINATION	-0.18983971	47	6.914e-06	3.398e-03	HMC15:41 HMC14:67 HMC14:342 HMC14:218 HMC14:273 H2BC15:313 H3C13:342
EROID_HORMONE_BIOSYNTHESIS	0.24043031	28	1.083e-05	3.398e-03	CYP17A1:47 HSD11B1:49 COMT:65 HSD17B2:124 CYP11A1:59 HSD17B3:195 SULT
E_SPERM_MOTILITY_AND_TAXES	0.42588855	8	3.036e-05	5.598e-03	KCNU1:286 CTSPPERD:403 CTSPPERSB:61 CTSPPERG:615 CTSPPSER:1018 CTSPPERA:1064
OME_BIOLOGICAL_OXIDATIONS	0.10075978	145	3.066e-05	5.598e-03	SULT1A2:14 SULT1A3:24 SULT1A4:28 MGST3:51 COMT:65 CYP27L1:75 CYP51A1
OME_SIGNALING_BY_GPCR	-0.05076419	595	3.415e-05	5.598e-03	CAMK2B:18 CDC42:49 TACR2:64 FSHR:125 SSTR4:127 RHOA:128 TIAM2:139
_ALPHA_12_13_SIGNALLING_EVENTS	-0.15513436	62	2.481e-05	5.598e-03	RHOA:128 TIAM2:139 ARHGEF1:145 ARHGEF2:173 OBSCN:396 FGD1:437 ARHGE
ETABOLIC_DISORDERS_OF_BIOLOGIC	0.21880596	30	3.408e-05	5.598e-03	CYP17A1:47 CYP21A2:380 CYP11A1:419 GTGL:457 CYP19A1:597 ACY1:708 CYP26
REACTOME_MEIOSIS	-0.13678867	75	4.386e-05	5.649e-03	UBE2I:7 HMC15:41 HMC14:67 HMC14:344 SYNE1:187 HMC14:218 HMC15:273
TRANSPORT_OF_INORGANIC_CATIONS	0.11833267	101	4.203e-05	5.649e-03	SLC38A4:57 SLC6A6:70 SLC26A3:82 SLC5A1:92 SLC3A1:111 SLC12A1:112 SLC1
_THROUGH_HOMOLOGOUS_RECOMBIN	-0.15106528	62	4.027e-05	5.649e-03	RPS27A:48 UBA52:88 POLD2:168 EME1:223 RTEL1:277 POLE:341 BLM:354
OME_GENE_SILENCING_BY_RNA	-0.12855410	82	5.975e-05	7.182e-03	POLR2J3:32 HMC15:41 RAN:55 HMC14:67 HMC14:344 NUP88:177 HMC2:118
E_ASPARTATE_AND_GLUTAMATE_MET	0.20492863	30	1.038e-04	1.170e-02	GLS2:36 GGT1:25 AXT1:103 CAD:354 CPS1:590 COT1:404 ASL:1005
THROCYTES_TAKE_UP_OXYGEN_AND	-0.05575982	6	1.108e-04	1.175e-02	CA2:106 AQP1:223 CA4:235 SLC3A1:382 RHAG:1001 CA1:1527 NA
AGE_SIGNALING_DEATH_TRANSDUKT	0.15134762	54	1.226e-04	1.185e-02	TIAM2:139 ARHGEF1:145 ARHGEF2:173 OBSCN:396 FGD1:437 ARHGE12:612 TI
RNA_DOUBLE_STRAND_BREAK_REPAIR	-0.09675813	133	1.249e-04	1.185e-02	UBE2I:7 HMC15:41 RPS27A:48 HMC14:67 UBA52:88 HMC14:344 POLD2:188
OME_HOMOLOGY_DIRECTED_REPAIR	-0.10712008	107	1.361e-04	1.227e-02	UBE2I:7 HMC15:41 RPS27A:48 HMC14:67 UBA52:88 HMC14:344 POLD2:188
S_NTR_RECEPTOR_MEDIATED_SIGNA	-0.11630786	85	2.185e-04	1.876e-02	RPS27A:48 UBA52:88 TIAM2:139 ARHGEF1:145 ARHGEF2:173 CASP3:226 OBSC
CELL_DEATH_SIGNALLING_VIA_NRAGE	-0.30489214	66	2.537e-04	2.079e-02	RPS27A:48 UBA52:88 TIAM2:139 ARHGEF1:145 ARHGEF2:173 CASP3:226 OBSC
MINO_ACID_TRANSPORT_ACROSS_THE	0.18884722	31	2.775e-04	2.097e-02	SLC38A4:57 SLC6A6:70 SLC36A1:11 SLC7A9:15 SLC6A19:173 SLC6A12:48 SLC5
THE_NEGATIVELY_REGULATES_RRNA	-0.18870388	31	2.801e-04	2.097e-02	HMC15:41 HMC14:67 HMC14:344 HMC14:218 H3C13:273 H2BC15:313 H3C13:342
SENSORY_PROCESSING_OF_SOUND	-0.12733561	68	2.908e-04	2.097e-02	SNAP25:11 PCDH15:13 ROPR2:129 CTBP2:252 CTG1:367 CACNB2:368 DNAT1:704
INE_AND_METHIONINE_METABOLISM	0.18908808	30	3.417e-04	2.369e-02	SOD5:37 TAT:54 GGT2:55 DMSH:364 DNMT1:265 CTH44:602 GGT1:403
E_SERINE_AND_THREONINE_METABO	0.08599666	31	3.578e-04	2.389e-02	SOD5:37 AGXT:103 GNM1:121 CHD1:39 DAP2:383 CTH44:602 PSPH:444
OXIDATIVE_PHOSPHORYLATION	0.10677390	92	3.723e-04	2.398e-02	NDUFS5:78 NDUFB5:486 ATP6V0A4:516 NDUFA9:602 COXSB:633 NDUFB8:636 NDU
ETABOLISM_OF_STEROID_HORMONES	0.18630625	30	4.175e-04	2.509e-02	CYP17A1:47 HSD11B1:49 HSD17B2:124 HSD17B3:195 AKR1B1:369 CYP21A2:380 CYP
E_DEATH_RECEPTOR_SIGNALING	-0.08782126	137	4.106e-04	2.509e-02	UBE2D1:5 RACK1:23 RPS27A:48 BIRC5:73 SLC6A19:173 SLC6A12:48 SLC5
GLYCOSYLATION_OF_TSR_DOMAIN_C	-0.16500465	38	4.352e-04	2.531e-02	HMC15:41 RPS27A:48 HMC14:67 UBA52:88 HMC14:344 HMC14:218 H3C13:273
ME_AMYLOID_FIBRIL_FORMATION	-0.13278525	58	4.803e-04	2.706e-02	HMC14:67 HMC14:344 HMC14:218 H3C13:273 H2BC15:313 H3C13:342
CTIVATED_PKB1_STIMULATES_TRANS	-0.19373075	27	4.983e-04	2.722e-02	ELOVL7:97 CYP11A1:59 HSD17B3:195 CP7IA:290 PTGS2:288 HMGCS2:293 SCD
OME_FATTY_ACID_METABOLISM	0.08151227	154	5.137e-04	2.724e-02	HMC14:67 HMC14:344 HMC14:218 H3C13:273 H2BC15:313 H3C13:342
ANOHMOLOGOUS_END_JOINING_NHEJ	-0.15396334	42	6.645e-04	2.908e-02	MAOGH:20 EFT1:31 POLR2J3:32 HMC15:41 ELOVL8:43 RPS27A:48 CDC42:49
OME_DEVELOPMENTAL_BIOLOGY	-0.36907818	792	5.936e-04	2.973e-02	PISD:109 SLC44A4:184 MIGA1:376 SLC44A4:247 GPCPD:413 PLGA218:441 DGAT
YCEROPHOSPHOLIPID_BIOSYNTHESI	0.09293836	114	6.372e-04	3.098e-02	IQGAP3:1 ACT1:29 CDC42:49 CPSF2:70 RHOA:128 NHS:134 TIAM2:139
TOME_RHO_GTPASE_CYCLE	-0.05058453	396	6.529e-04	3.098e-02	SNAP25:11 CAMK2B:18 SLITRK4:87 LINC7:207 RHOA:128 NHS:134 KCVN2:232 GRINR
OTOME_NEURONAL_SYSTEM	-0.05231558	352	8.545e-04	3.845e-02	PMS2:91 BLM:354 TOP3A:662 ATM:952 NEIL1:1112 RAD50:1364 HUS1:1431
OME_DISEASES_OF_DNA_REPAIR	-0.13955095	48	8.377e-04	3.845e-02	

MGI_Mammalian_Phenotype_Level_4 Top pathways by non-permulation